

Polar Coordinates and Conversions



Converting points

- Convert each polar point (r, θ) to rectangular coordinates (exact values):
a $(4, \frac{\pi}{3})$ **b** $(2, \frac{3\pi}{4})$
 - Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:
a $(-3, 3)$ **b** $(0, -5)$
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Converting equations

- Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.
 - Convert each rectangular equation to polar form:
a $x^2 + y^2 = 16$ **b** $y = x$
 - Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.
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Multiple representations

- Give two other pairs (r, θ) that represent the same point as $(3, \frac{\pi}{4})$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.
 - Plot-reasoning: explain why the polar point $(-2, \frac{\pi}{2})$ corresponds to the same location as $(2, \frac{3\pi}{2})$.
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Distance between polar points

- Convert $(4, \frac{\pi}{6})$ and $(4, \frac{2\pi}{3})$ to rectangular coordinates, then find the distance between them, to two decimal places.
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Further conversion practice

- Convert each polar point to rectangular coordinates (exact values):
a $(6, \pi)$ **b** $(5, \frac{5\pi}{6})$

- 10 Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.
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Why polar coordinates are not unique

- 11 Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.
- 12 The pole (origin) is a special case in polar coordinates. Explain why.
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Converting more complex equations

- 13 Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.
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More point conversions with negative r

- 14 Convert the polar point $(-4, \frac{\pi}{6})$ (negative r) to rectangular coordinates.
- 15 Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.
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Converting a spiral equation

- 16 The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.
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Converting a real-world radar equation

- 17 A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.
- 18 A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \leq \theta < 2\pi$, to two decimal places.
- 19 Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r .

- 20** Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.